

$$2y^4 \frac{\partial z}{\partial x} - xy \frac{\partial z}{\partial y} = x\sqrt{z^2 + 1} \cdot \frac{2y^4}{x} \cdot \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = \sqrt{z^2 + 1}.$$

$$\frac{x dx}{2y^4} = \frac{dy}{-y} = \frac{dz}{\sqrt{z^2 + 1}}.$$

$$1) \quad x dx = -2y^3 dy, \quad x^2 + y^4 = A.$$

$$2) \quad \frac{dy}{-y} = \frac{dz}{\sqrt{z^2 + 1}}, \quad \operatorname{arsh} z + \ln|y| = B.$$

В результате $F(x^2 + y^4; \operatorname{arsh} z + \ln|y|) = 0$. $\operatorname{arsh} z + \ln|y| = f(x^2 + y^4)$.

$$z = \operatorname{sh}\left(f(x^2 + y^4) - \ln|y|\right).$$